

HBM_G0V60C15

Adaptation to 48V - 15A Class Device

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1. Introduction

This document outlines the engineering rationale and sizing process for adapting the HBM_G0VH4C5 motor drive from a 400V - 5A configuration to a new 48V - 15A architecture (HBM_G0V60C15).

To accommodate the higher current and lower voltage, the power stage and control circuitry have been redesigned. This includes selecting appropriate MOSFETs (evaluating HGN036N08S), updating the Motor Gate Driver (evaluating EG2131), and recalculating current/voltage sensing and snubber networks.

A strict design constraint is that all components must be sourced locally (e.g., UGE Electronics). This document serves as the standard reference for all mathematical equations, component sizing, and engineering justifications applied to this redesign.

1.1 MOSFET

The selected MOSFET for the 48V / 15A power stage is the **IRFB4110PbF** from Infineon Technologies (International Rectifier). The device was selected to match the updated system voltage and current specifications, significantly minimizing conduction losses, providing robust safety margins against voltage spikes, and ensuring high thermal and electrical reliability during motor driving operations.

Datasheet: [IRFB4110PbF](#)

Selection rationale:

- **Local availability and cost:** The IRFB4110 is an industry-standard power MOSFET that is widely stocked and readily available from local suppliers at a competitive cost, making it highly practical for prototyping, assembly, maintenance, and replacement.
- **100 V drain-source rating (VDS):** The 100 V maximum VDS rating provides a $>2\times$ safety margin over the nominal 48 V DC-link voltage. This extensive margin offers superior protection against inductive switching spikes, ringing across the half-bridge nodes, and transient voltage surges during regenerative braking.
- **High continuous current capability (ID):** Rated for up to 180 A (silicon-limited) and 120 A (package-limited at $T_C=25^\circ\text{C}$), the MOSFET provides more than $8\times$ headroom over the nominal 15 A operating current. This substantial margin easily handles high peak phase currents, motor startup inrush, and locked-rotor conditions without entering thermal runaway.
- **Ultra-low on-state resistance (RDS(on)):** With a typical RDS(on) of 3.7 m Ω (max 4.5 m Ω at VGS=10 V), the device dramatically slashes I²R conduction losses compared to higher-voltage alternatives, ensuring high power conversion efficiency and reducing heatsink size requirements.
- **Low thermal resistance and TO-220 package:** The TO-220AB package features a low junction-to-case thermal resistance ($R_{\theta JC}\approx 0.40^\circ\text{C/W}$), ensuring efficient heat transfer to the heatsink and simplified mechanical mounting.
- **Gate charge and driver compatibility:** With a total gate charge (Qg) of approximately 150 nC (and Miller charge Qgd $\approx$ 53 nC), the MOSFET achieves an optimal balance between fast switching transitions (reducing switching losses) and manageable peak drive currents for the dedicated gate driver at the target PWM frequency.
- **Avalanche and body-diode ruggedness:** The device features high repetitive avalanche capability (EAS) and a fast-recovery integral body diode, effectively handling inductive energy recirculation and freewheeling currents during inverter dead-time.

1.2 Gate Drive

The selected gate-driver IC for the 48 V / 15 A power stage is the **EG2131** from EGmicro. The device was selected to meet the switching requirements of the IRFB4110 MOSFETs, provide high noise immunity, and deliver reliable high-speed gate driving for the updated half-bridge power stage.

Datasheet: [EG2131](#)

Selection Rationale:

- **Local availability and cost:** The EG2131 is widely available from local suppliers at a competitive unit cost, making it a practical choice for prototyping, PCB assembly, maintenance, and replacement.
- **High-side floating channel capability:** The high-voltage section is rated for operation up to 300 V, providing more than a 6× safety margin over the nominal 48 V DC-link voltage. This additional voltage margin improves immunity to inductive phase-node ringing and transient voltage spikes during switching.
- **High peak drive current:** The driver provides high output drive capability, with a typical source current of 1.0 A and sink current of 1.5 A. This allows fast charging and discharging of the IRFB4110 gate, which has a total gate charge of approximately 150 nC. As a result, the turn-on and turn-off transition times are reduced, helping to minimize dynamic switching losses.
- **Integrated interlock and hardware dead time:** The driver includes built-in cross-conduction prevention logic and internal dead-time generation. This hardware interlock prevents the high-side and low-side MOSFETs from turning on simultaneously, protecting the half-bridge against shoot-through conditions and potential device failure.
- **External bootstrap diode configuration:** Instead of relying on an integrated bootstrap diode, the design uses a dedicated external ultrafast-recovery diode. This configuration reduces thermal dissipation within the driver IC, minimizes forward-recovery voltage effects, and improves bootstrap capacitor recharge during high-frequency PWM operation.
- **Compact SOIC-8 footprint:** The standard SOP-8 package allows the driver to be placed close to the power MOSFETs. This shortens the gate-drive paths, reduces parasitic inductance in the gate loop, and helps suppress ringing caused by high di/dt switching currents.

1.3 External Components

The external components for the gate-drive circuitry are sized according to the electrical requirements of the 48 V / 15 A power stage, the switching dynamics of the IRFB4110 MOSFET, and the drive capability of the EG2131 gate driver.

1.3.1 Gate Resistor and Switching Loss Calculations

Two distinct analytical models are evaluated to determine the gate resistor sizing and switching loss estimation: the Slew-Rate Target Method (First-Order Model) and the Piecewise Gate-Charge Model (Physics-Based Crossover Model).

Method 1: Slew-Rate Based Target Estimation (First-Order Approximation)

This approach treats switching transitions as design targets defined by acceptable switch-node dv/dt rates to mitigate electromagnetic interference (EMI) and phase-node voltage overshoot.

1. Derivation of Target Switching Times:

For a nominal $V_{DS} = 48$ V, target voltage slew rates are established:

- Turn-on target slew rate: ≤ 1.2 V/ns
- Turn-off target slew rate: ≤ 1.6 V/ns

The resulting target switching durations are:

$$t_{on} \approx \frac{V_{DS}}{\text{Slew Rate}_{on}} = \frac{48V}{1.2 \text{ V/ns}} \approx 40 \text{ ns}$$

$$t_{off} \approx \frac{V_{DS}}{\text{Slew Rate}_{off}} = \frac{48V}{1.6 \text{ V/ns}} \approx 30 \text{ ns}$$

2. First-Order Switching Loss Calculation:

Assuming linear simultaneous transitions of drain voltage and drain current across the full duration:

$$P_{SW}(1st) \approx \frac{1}{2} V_{DS} I_D (t_{on} + t_{off}) f_{SW}$$

For $V_{DS} = 48$ V, $I_D = 15$ A, and $f_{sw} = 20$ kHz:

$$P_{SW} \approx 0.5 \times 48 \times 15 \times (30+40) \times 10^{-9} \times 10^3 = 0.504 \text{ W}$$

This is a first-order estimate of the voltage-current overlap loss and does not include other switching-related losses such as EOSS, body-diode reverse-recovery losses, gate-drive losses, or additional losses associated with switch-node ringing.

Method 2: Piecewise Gate-Charge & Miller Plateau Analysis (Rigorous Model)

This method models the actual physical charge transfer into the gate capacitances (C_{gs} and C_{gd}) using datasheet charge curves and gate loop impedances.

1. Extracted Device Parameters (IRFB4110 & EG2131):

- Gate drive supply voltage: $V_{CC} = 12\text{ V}$
- Gate plateau voltage at $I_D = 15\text{ A}$: $V_{plat} \approx 5.0\text{ V}$
- Gate threshold voltage: $V_{th} \approx 3.7\text{ V}$
- Gate-to-drain (Miller) charge: $Q_{gd} \approx 53\text{ nC}$
- Post-threshold gate-to-source charge: $Q_{gs2} = Q_{gs} \times (1 - (V_{th} / V_{plat})) \approx 36\text{ nC} \times (1 - (3.7 / 5.0)) \approx 9.36\text{ nC}$
- Output capacitance charge (48 V): $Q_{oss} \approx 80\text{ nC}$ ($C_{oss,eff} \approx 1.67\text{ nF}$)
- Reverse recovery charge: $Q_{rr} \approx 160\text{ nC}$
- Total gate loop resistance (using an external $R_{G,ext} = 10\ \Omega$):
 - Internal MOSFET gate resistance: $R_{G,int} \approx 1.2\ \Omega$
 - Driver internal pull-up resistance: $R_{drv,on} \approx 10\ \Omega$
 - Driver internal pull-down resistance: $R_{drv,off} \approx 5.3\ \Omega$
 - $R_{tot,on} = R_{drv,on} + R_{G,ext} + R_{G,int} = 10 + 10 + 1.2 = 21.2\ \Omega$
 - $R_{tot,off} = R_{drv,off} + R_{G,ext} + R_{G,int} = 5.3 + 10 + 1.2 = 16.5\ \Omega$

2. Dynamic Drive Current and Crossover Timing:

During active switching, current and voltage overlap only during the current rise/fall $\left(\frac{t_{ri}}{t_{fi}}\right)$ and voltage fall/rise $\left(\frac{t_{fv}}{t_{rv}}\right)$ phases:

Turn-On Transition:

$$I_{G,on(plat)} = \frac{(V_{CC} - V_{plat})}{R_{tot,on}}$$

$$= \frac{(12\text{ V} - 5.0\text{ V})}{21.2\ \Omega} \approx 0.33\text{ A}$$

$$t_{ri} = \frac{Q_{gs2}}{I_{G,on(plat)}} = \frac{9.36\text{ nC}}{0.33\text{ A}} \approx 28.4\text{ ns (Current Rise Time)}$$

$$t_{fv} = \frac{Q_{gd}}{I_{G,on(plat)}} = \frac{53\text{ nC}}{0.33\text{ A}} \approx 160.6\text{ ns (Voltage Fall Time)}$$

$$t_{cross,on} = t_{ri} + t_{fv} = 28.4\text{ ns} + 160.6\text{ ns} = 189.0\text{ ns}$$

Turn-Off Transition:

$$I_{G,off(plat)} = \frac{V_{plat}}{R_{tot,off}} = 5.0\text{ V} / 16.5\ \Omega \approx 0.303\text{ A}$$

$$t_{rv} = \frac{Q_{gd}}{I_{G,off}(plat)} = \frac{53 \text{ nC}}{0.303 \text{ A}} \approx 174.9 \text{ ns (Voltage Rise Time)}$$

$$t_{fi} = \frac{Q_{gs2}}{I_{G,off}(plat)} = \frac{9.36 \text{ nC}}{0.303 \text{ A}} \approx 30.9 \text{ ns (Current Fall Time)}$$

$$t_{cross,off} = t_{rv} + t_{fi} = 174.9 \text{ ns} + 30.9 \text{ ns} = 205.8 \text{ ns}$$

3. Comprehensive Switching Loss Formulation:

The total dynamic power dissipation per MOSFET accounts for crossover losses, capacitive energy discharge, and body-diode recovery:

$$P_{cross} = 0.5 \times V_{DS} \times I_D \times (t_{cross.on} + t_{cross.off}) \times f_{sw}$$

$$P_{cross} = 0.5 \times 48 \times 15 \times (189.0 + 205.8) \times 10^{-9} \times 20 \times 10^3 \approx 2.84 \text{ W}$$

$$P_{oss} = 0.5 \times C_{oss,eff} \times (V_{DS})^2 \times f_{sw}$$

$$P_{oss} = 0.5 \times 1.67 \times 10^{-9} \times (48)^2 \times 20 \times 10^3 \approx 0.038 \text{ W}$$

$$P_{rr} = Q_{rr} \times V_{DS} \times f_{sw}$$

$$P_{rr} = 160 \times 10^{-9} \times 48 \times 20 \times 10^3 \approx 0.154 \text{ W}$$

$$P_{SW,total} = P_{cross} + P_{oss} + P_{rr} = 2.84 + 0.038 + 0.154 \approx 3.03 \text{ W}$$

Comparative Evaluation: Why Method 2 is More Accurate

Evaluation Criterion	Method 1: Slew-Rate / Target Model	Method 2: Piecewise Gate-Charge Model
Physical Basis	Empirical assumption based on arbitrary dv/dt limits	Electrostatic charge conservation (Q = Integral of I dt) across non-linear junctions
Miller Effect	Ignored; assumes linear transition through the entire turn-on window	Directly modeled via Q_gd and Miller Plateau voltage clamp (V_plat)

Evaluation Criterion	Method 1: Slew-Rate / Target Model	Method 2: Piecewise Gate-Charge Model
Gate Impedance	Ignores driver pull-up/down resistance and internal silicon gate resistance	Models total loop resistance ($R_{drv} + R_{G,ext} + R_{G,int}$)
Delay Discrimination	Blurs turn-on delay ($t_{d(on)}$) with true power crossover (t_{cross})	Separates zero-loss turn-on delay from active $V \times I$ overlap
Secondary Losses	Excludes E_{oss} dissipation and body diode reverse recovery (Q_{rr})	Integrates P_{cross} , P_{oss} , and P_{rr} into total switching dissipation

Technical Conclusion:

Method 1 serves solely as an initial rule-of-thumb to set boundary limits on transition times. Method 2 is vastly more accurate because it calculates the actual drive current available at the Miller plateau (V_{plat}), capturing the dominant voltage transition phase (t_{fv} , t_{rv}) caused by the high non-linear transfer capacitance (C_{rss}). Consequently, Method 2 prevents significant underestimation of switching losses, ensuring reliable thermal margin calculation and proper heatsink sizing.

4. Gate Impedance Network and Capacitance Elimination

In traditional low-speed switching or poorly laid out gate loops, designers sometimes attempt to add an external gate-to-source capacitor ($C_{gs,ext}$) or parallel RC snubber directly across the gate and source terminals to filter out switching noise, damp high-frequency ringing, and prevent unintended dv/dt turn-on. In this updated 48 V / 15 A power stage, external gate-to-source capacitance is strictly omitted from the design.

The Flaw of Increasing Gate Capacitance:

While adding external gate-to-source capacitance may seem like an intuitive way to suppress gate ringing and clamp the gate voltage below the threshold (V_{th}), it introduces severe detrimental trade-offs:

- **Overloading Driver Current Capability:** The EG2131 gate driver features a peak source current of 1.0 A and sink current of 1.5 A. Adding external capacitance drastically increases the total charge:

$$Q_{total} = Q_g + (C_{ext} \times V_{GS})$$

This excessive charge must be delivered and removed every switching cycle, driving the output stages of the IC into prolonged saturation.

- **Severe Gate Driver Thermal Dissipation:** The power dissipated inside the gate driver IC scales directly with total capacitance:

$$P_{driver} = Q_{total} \times V_{DRV} \times f_{sw} = (Q_g + C_{ext} \times V_{DRV}) \times V_{DRV} \times f_{sw}$$

Adding even a few nanofarads of external capacitance multiplies the driver package power dissipation, causing thermal runaway in the compact SOIC-8 package at 20 kHz operation.

- **Massive Increase in Switching Losses:** Artificially increasing gate capacitance elongates the gate voltage rise time and prolongs the Miller plateau transition (t_{Miller}). Because the MOSFET remains in the active linear region for a significantly longer duration, the voltage-current crossover loss (P_{cross}) escalates dramatically, overheating the MOSFETs.
- **Deadtime and Timing Skew Degradation:** The extra capacitive charge introduces large turn-on and turn-off propagation delays, compromising the fixed hardware deadtime (typical 300–400 ns) and creating a dangerous risk of half-bridge shoot-through.

Correct Engineering Implementation:

Ringing and noise immunity are instead controlled purely by sizing a single, optimized series gate damping resistor ($R_{GateExternal}$) and minimizing the physical PCB gate-drive loop area to reduce parasitic inductance.

5. Gate Current and Gate Resistor Sizing

The gate current required during the Miller transition is defined by the MOSFET gate-to-drain Miller charge (Q_{Miller} or Q_{gd}) and the desired transition time:

$$I_{\text{G}} \approx Q_{\text{Miller}} / t_{\text{Miller}}$$

For the selected **IRFB4110PbF**:

$$Q_{\text{Miller}} \approx 53 \text{ nC}$$

Turn-On Transition

The effective gate charging current during the Miller plateau is governed by the gate-drive supply voltage, the plateau voltage, and the total turn-on loop resistance:

$$I_{\text{G,on}} = (V_{\text{DRV}} - V_{\text{PL}}) / (R_{\text{GateInternal}} + R_{\text{GateDriverSource}} + R_{\text{GateExternal}})$$

Using:

- $V_{\text{DRV}} = 12 \text{ V}$ (Gate driver supply voltage)
- $V_{\text{PL}} \approx 5.0 \text{ V}$ (MOSFET Miller plateau voltage at $I_{\text{D}} = 15 \text{ A}$)
- $R_{\text{GateInternal}} = 1.2 \text{ } \Omega$ (IRFB4110 intrinsic silicon gate resistance)
- $R_{\text{GateDriverSource}} \approx 10.0 \text{ } \Omega$ (EG2131 internal P-channel pull-up equivalent resistance)
- $R_{\text{GateExternal}} = 10.0 \text{ } \Omega$ (Selected external series damping resistor)

$$I_{\text{G,on}} = \frac{(12 \text{ V} - 5.0 \text{ V})}{(1.2 \text{ } \Omega + 10.0 \text{ } \Omega + 10.0 \text{ } \Omega)} = \frac{7.0 \text{ V}}{21.2 \text{ } \Omega} \approx 0.33 \text{ A (330 mA)}$$

The resulting turn-on Miller plateau transition duration is:

$$t_{\text{Miller,on}} = \frac{Q_{\text{Miller}}}{I_{\text{G,on}}} = \frac{53 \text{ nC}}{0.33 \text{ A}} \approx 160.6 \text{ ns}$$

Turn-Off Transition

During turn-off, the gate discharges from the plateau voltage down to ground through the turn-off path impedance:

$$I_{\text{G,off}} = \frac{V_{\text{PL}}}{(R_{\text{GateInternal}} + R_{\text{GateDriverSink}} + R_{\text{GateExternal}})}$$

Using:

- $V_{\text{PL}} \approx 5.0 \text{ V}$
- $R_{\text{GateInternal}} = 1.2 \text{ } \Omega$
- $R_{\text{GateDriverSink}} \approx 5.3 \text{ } \Omega$ (EG2131 internal N-channel pull-down equivalent resistance)

- $R_{_GateExternal} = 10.0 \, \Omega$

$$I_{_G,off} = \frac{5.0 \, V}{(1.2 \, \Omega + 5.3 \, \Omega + 10.0 \, \Omega)} = \frac{5.0 \, V}{16.5 \, \Omega} \approx 0.303 \, A \, (303 \, mA)$$

The resulting turn-off Miller plateau transition duration is:

$$t_{_Miller,off} = \frac{Q_{Miller}}{I_{_G,off}} = \frac{53 \, nC}{0.303 \, A} \approx 174.9 \, ns$$

Sizing Evaluation and Driver Asymmetry

The EG2131 internal output stage exhibits inherent impedance asymmetry ($R_{_GateDriverSink} \approx 5.3 \, \Omega$ vs. $R_{_GateDriverSource} \approx 10.0 \, \Omega$). Because the discharge voltage across the gate path during turn-off is limited to $V_{_PL} \approx 5.0 \, V$ (compared to $V_{_DRV} - V_{_PL} = 7.0 \, V$ during turn-on), the lower sink impedance naturally balances the peak discharge current ($I_{_G,off} \approx 303 \, mA$) with the charge current ($I_{_G,on} \approx 330 \, mA$).

A single $10 \, \Omega$ external resistor provides an optimal balance:

- It damps parasitic LC gate oscillations without exceeding the 1.0 A / 1.5 A current limits of the driver.
- It maintains a well-controlled transition time (≈ 160 – $175 \, ns$) that keeps switching losses low while preventing severe phase-node voltage spikes and false dv/dt induced turn-on

1.3.2 Bootstrap Capacitor

The bootstrap capacitor provides the local energy reservoir and low-impedance current path required to drive the high-side MOSFET. Because the high-side gate is charged through this capacitor, its value must be sufficiently large to maintain the bootstrap supply above the gate driver's high-side Under-Voltage Lockout (UVLO) threshold throughout the switching cycle.

A simple design rule is to select the bootstrap capacitance as at least ten times the effective gate capacitance of the MOSFET. As a general rule of thumb, the capacitor should store sufficient charge to drive the high-side MOSFET without its voltage being depleted by more than approximately 10% during a switching cycle:

$$C_{_BOOT} \geq 10 \times C_{_G}$$

where the effective gate capacitance can be approximated from the MOSFET total gate charge:

$$C_{_G} = \frac{Q_G}{V_{Q1p}}$$

For the selected **IRFB4110PbF**:

$$Q_G \approx 150 \text{ nC}$$

and the target gate-drive voltage is:

$$V_{Q1p} \approx V_{GS} = 12 \text{ V}$$

Therefore:

$$C_G \approx \frac{150 \text{ nC}}{12 \text{ V}} = 12.5 \text{ nF}$$

and the simple 10× rule gives:

$$C_{BOOT} \geq 125 \text{ nF (0.125 } \mu\text{F)}$$

This provides a useful lower-bound estimate; however, the actual bootstrap capacitor should also account for MOSFET gate charge, gate-driver quiescent and level-shifter operating currents, maximum PWM duty cycle, bootstrap diode forward voltage drop, ceramic capacitor DC-bias capacitance derating, temperature variations, and the allowable bootstrap voltage ripple.

A substantially larger **2.2 μF (50 V, X7R ceramic)** bootstrap capacitor was selected for the power stage.

Justification:

The larger capacitor was intentionally selected to provide substantial charge reserve rather than minimizing the capacitance to the theoretical requirement. The bootstrap supply is fundamentally a charge-balance system, and the capacitor voltage decreases according to the charge removed from it:

$$\Delta V_{BOOT} = \frac{\Delta Q}{C_{BOOT}}$$

For a single high-side switching event, the MOSFET requires approximately:

$$Q_G \approx 150 \text{ nC}$$

Ignoring the additional driver leakage and quiescent-current contributions, the corresponding voltage drop of the selected 2.2 μF capacitor is only:

$$\Delta V_{BOOT} \approx \frac{150 \text{ nC}}{2.2 \mu\text{F}} \approx 68.2 \text{ mV}$$

Therefore, even over ten consecutive high-side switching events without a significant bootstrap refresh:

$$\Delta V_{BOOT,10} \approx 10 \times 68.2 \text{ mV} = 682 \text{ mV (0.682 V)}$$

The resulting small voltage droop provides substantial tolerance to skipped or shortened bootstrap recharge intervals (such as during transient high duty-cycle operation or low-side freewheeling deadtimes) and allows the EG2131 high-side driver to maintain a stable gate-drive voltage well above its UVLO threshold (typical 7.5 V – 8.5 V).

The larger capacitance also provides increased immunity to variations in MOSFET gate charge, gate-driver consumption, capacitor tolerance, temperature, and DC-bias derating (where effective capacitance drops by ~30% at 12 V bias). The bootstrap capacitor therefore has considerable charge reserve relative to the approximately 150 nC gate-charge requirement.

The principal trade-off is the increased charge required to initially charge the bootstrap capacitor and initial charge time to 90%:

$$0.9 = \left(1 - e^{-\frac{t}{\tau}}\right)$$

$$\tau = R_BOOT_path \times C_BOOT = 10 \, \Omega \times 2.2 \, \mu F = 22 \, \mu s$$

$$t \approx 2.3 \times \tau = 2.3 \times 22 \, \mu s \approx 50.6 \, \mu s$$

The bootstrap charging path, including the external ultrafast bootstrap diode and series limiting resistor (total loop resistance $\approx 10 \, \Omega$), easily accommodates the associated charging current. This initial charging duration of approximately 50.6 μs is negligible compared to the inverter pre-charge and initialization routines, while the resulting increase in bootstrap hold-up capability provides a critical robustness margin.

The selected value should consequently be understood as a **hold-up and robustness choice**, rather than a minimum capacitance requirement.

1.3.3 VCC Decoupling Capacitor

The gate-driver supply requires a local decoupling capacitor to provide the high-frequency current demanded by the gate-drive output stage and to minimize voltage sag and supply-loop inductance during switching transitions.

As an initial design rule, the local VCC decoupling capacitance is selected as:

$$C_VCC \geq 10 \times C_BOOT = 10 \times 2.2 \, \mu F = 22 \, \mu F$$

This provides a substantially larger local energy reservoir than the bootstrap capacitor and helps maintain a stable gate-driver supply during high-current switching transitions.

To avoid anti-resonance impedance peaks and parasitic LC ringing that often arise from placing mismatched capacitor values in parallel, a single uniform low-ESR ceramic capacitor value (22 μF , 25 V, X7R MLCC) is selected.

The decoupling capacitor should be placed directly adjacent to the **EG2131** VCC and GND pins, minimizing the associated PCB loop area and parasitic trace inductance.

1.3. RC Snubber

A first-pass RC snubber value was estimated from the effective high-frequency switching-loop inductance and the MOSFET parasitic output capacitance. The initial values are:

$$C_{\text{snub}} \approx 4 \times C_{\text{oss}} \approx 4 \times 680 \text{ pF} \approx 2.7 \text{ nF}$$

$$R_{\text{snub}} \approx \sqrt{\frac{L_0}{C_{\text{oss}}}} \approx \sqrt{\frac{15 \text{ Nh}}{680 \text{ Pf}}} \approx 4.7 \text{ } \Omega$$

where:

- $C_{\text{oss}} \approx 680 \text{ pF}$ represents the high-voltage drain-to-source capacitance of the **IRFB4110PbF** at the nominal operating voltage ($V_{\text{DS}} = 48 \text{ V}$).
- $L_0 \approx 15 \text{ nH}$ represents the total estimated high-frequency switching-loop inductance, combining PCB power loop stray inductance ($L_{\text{Power}} \approx 10 \text{ nH}$) and TO-220 internal package/lead inductance ($L_{\text{Package}} \approx 5 \text{ nH}$).

Snubber Power Dissipation Estimate

The continuous power dissipated in the snubber resistor across the 48 V DC-link at a switching frequency of $f_{\text{sw}} = 20 \text{ kHz}$ is calculated as:

$$P_{\text{snub}} \approx C_{\text{snub}} \times (V_{\text{DS}})^2 \times f_{\text{sw}}$$

$$P_{\text{snub}} \approx 2.7 \times 10^{-9} \text{ F} \times (48 \text{ V})^2 \times 20 \times 10^3 \text{ Hz}$$

$$P_{\text{snub}} \approx 2.7 \times 10^{-9} \times 2304 \times 20000 \approx 0.125 \text{ W}$$

Design Assessment:

These values establish a near-critically damped termination network for the switch-node parasitic LC tank, effectively suppressing high-frequency voltage ringing and clamping peak inductive overshoots without causing excessive power loss (a standard 0.25 W or 0.5 W SMD resistor, such as a 1206 or 1210 package, provides ample thermal margin).

The final snubber values are to be verified and fine-tuned experimentally or through transient switching-waveform simulation by evaluating the resulting switch-node overshoot, damping ratio, overall switching efficiency, and snubber thermal dissipation.

1.4 Current Sensing

Since galvanic isolation is not enforced between the power and logic domains, low-side shunt current sensing is employed to provide a cost-effective, high-accuracy, and compact current measurement solution. The shunt is placed in the low-side current return path of each half-bridge, allowing the resulting differential voltage drop to be amplified with respect to the local logic ground.

The **INA199A1** bidirectional current-sense amplifier from Texas Instruments is employed for this function.

Datasheet: INA199A1

Selection rationale:

- **Local availability and cost:** The device is widely stocked and readily available from local component distributors at a low unit cost, making it highly practical for prototyping, production, and field maintenance.
- **Common-mode voltage range:** The amplifier operates over a common-mode voltage range of -0.1 V to 26 V independent of the supply voltage, easily accommodating inductive ground bounce and negative switching transients occurring on the low-side source node.
- **High CMRR:** A typical common-mode rejection ratio (CMRR) of 120 dB provides exceptional rejection of high-frequency common-mode noise and switching transients appearing across the shunt.
- **Precision accuracy and low offset:** Features a low typical gain error of $\pm 0.5\%$ (max $\pm 1\%$) and an ultra-low input offset voltage ($V_{OS} \leq \pm 150 \mu V$), ensuring accurate current measurement for Field Oriented Control (FOC) without necessitating complex calibration routines.
- **Bandwidth:** The 80 kHz small-signal bandwidth is well-matched to the 20 kHz PWM fundamental frequency, providing clean tracking of phase currents while rejecting high-frequency switching hash.
- **High fixed gain:** The 50 V/V fixed precision gain allows the use of an ultra-low-value shunt resistor, minimizing I^2R power dissipation in the current path while delivering a robust signal swing for the 3.3 V microcontroller ADC.

Shunt Resistor Selection

The shunt resistance is selected as an optimal compromise between ADC dynamic range utilization, signal-to-noise ratio (SNR), and thermal power dissipation. For the updated **15 A nominal current** power stage, a low-inductance **2 m Ω metal-alloy shunt resistor** is selected.

To support bidirectional phase-current sensing (monitoring both motoring and regenerative braking currents), the reference pin of the amplifier is biased at the midpoint of the 3.3 V ADC range:

$$V_{\text{bias}} = \frac{3.3 \text{ V}}{2} = 1.65 \text{ V}$$

This establishes the zero-current operating point at exactly 1.65 V, enabling symmetric positive and negative current measurement.

For a 2 mΩ shunt and a fixed amplifier gain of 50 V/V, the sensed output voltage is:

$$V_{\text{signal}} = I \times 50 \times R_{\text{shunt}} + 1.65 \text{ V}$$

Substituting the selected shunt resistance ($R_{\text{shunt}} = 2 \text{ m}\Omega = 0.002 \text{ }\Omega$):

$$V_{\text{signal}} = I \times \left(50 \times 0.002 \frac{\text{V}}{\text{A}} \right) + 1.65 \text{ V}$$

Thus, the current-sensing path provides an overall transfer gain of $0.10 \frac{\text{V}}{\text{A}}$ ($100 \frac{\text{mV}}{\text{A}}$).

The theoretical 0 V to 3.3 V ADC input span corresponds to:

$$I_{\text{max}} = \frac{(3.3 \text{ V} - 1.65 \text{ V})}{0.10 \frac{\text{V}}{\text{A}}} = +16.5 \text{ A}$$

$$I_{\text{min}} = \frac{(0.0 \text{ V} - 1.65 \text{ V})}{0.10 \frac{\text{V}}{\text{A}}} = -16.5 \text{ A}$$

This yields a full-scale bidirectional measurement range of:

$$\text{Range} = \pm 16.5 \text{ A}$$

(Design Note: If higher transient peak current headroom is required during aggressive acceleration or motor stall conditions, a 1 mΩ shunt can be substituted, which doubles the measurable dynamic range to ±33 A with a sensitivity of 50 mV/A).

In practice, the usable linear range is slightly constrained by the amplifier output stage rail-to-rail swing limits (typically 30 mV to 50 mV from rails) and ADC conversion margins.

1.5 Voltage Sensing

For the 48 V / 15 A power stage, galvanic isolation is not enforced between the power and logic domains, and PGND and LGND are tied at a single star-ground point. This allows the switch-node (phase) voltage to be sensed directly with respect to LGND using a precision resistive voltage divider.

The divider ratio is selected such that the maximum expected switch-node voltage (including battery charge headroom and regenerative braking voltage rise up to 60 V) is mapped cleanly into the valid 0 to 3.3 V range of the microcontroller ADC:

$$V_{\text{signal}} = V_{\text{sw}} \times \left(\frac{3.3 \text{ k}}{(60 \text{ k} + 3.3 \text{ k})} \right)$$

where the upper leg is implemented as a series combination of three 20 kΩ SMD resistors (total 60 kΩ), and the lower leg is a 3.3 kΩ resistor.

This gives the following nominal full-scale mapping:

$$60 \text{ V} \rightarrow 3.128 \text{ V}$$

(Full-scale saturation limit: 63.3 V → 3.300 V)

The resulting scaling makes effective use of the available ADC dynamic range while retaining a safe margin below the 3.3 V rail during transient voltage overshoots. The upper divider is implemented as a series string of resistors to distribute voltage stress across multiple SMD footprints (0603 package size) and to prevent dielectric breakdown.

Power Dissipation

The power dissipated by the divider under the nominal 48 V DC switch-node condition is approximately:

$$P_{\text{divider}} = \frac{(48)^2}{(60 \text{ k} + 3.3 \text{ k})} = \frac{2304}{63300} \approx 0.0364 \text{ W (36.4 mW)}$$

This minimal power dissipation is distributed across the series resistor string rather than concentrated in a single component, resulting in negligible self-heating and minimal thermal drift error.

Low-Pass Filtering (Anti-Aliasing & Noise Rejection)

To suppress high-frequency PWM switching hash and switch-node ringing, a **47 nF ceramic capacitor** is placed in parallel with the 3.3 kΩ lower divider resistor (forming the `Phase_Cnd` conditioning node).

The equivalent Thevenin resistance seen by the filter capacitor is:

$$R_{\text{eq}} = 60 \text{ k}\Omega \parallel 3.3 \text{ k}\Omega = (60 \times 3.3) / (60 + 3.3) \approx 3.13 \text{ k}\Omega$$

The resulting low-pass filter cutoff frequency is:

$$f_c = \frac{1}{(2 \times \pi \times R_{eq} \times C_{filter})}$$

$$f_c = \frac{1}{(2 \times 3.1416 \times 3130 \, \Omega \times 47 \times 10^{-9} F)} \approx 1.08 \, kHz$$

This low cutoff frequency effectively attenuates the 20 kHz PWM carrier frequency and high-frequency switching harmonics, providing a clean, smoothed analog representation of the fundamental phase voltage for the ADC.

Operational Purpose:

The conditioned phase voltage signal (Phase_Cnd) is primarily utilized for:

- **High-Impedance Phase Monitoring:** Sensing the phase voltage when both high-side and low-side MOSFETs are turned OFF (tri-state), enabling Back-EMF zero-crossing detection for sensorless motor commutation and rotor position estimation.
- **Fault Diagnostics & Protection:** Real-time monitoring of phase-node state, open-phase detection, and regenerative bus over-voltage tracking during dynamic deceleration.